Multidimensional Difference Arrays

via Discrete Calculus

(https://github.com/mazmap/multidimensional-difference-arrays)

↑ Slides ↑

Who am I?

- Matteo Mertz

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- Finishing my bachelors degree in mathematics at TUM

Who am I?

- Matteo Mertz
- Finishing my bachelors degree in mathematics at TUM
- Currrently writing my bachelors thesis on the *Efficient computation of Euler characteristic profiles (of cubical filtrations of digital images)*

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- A runtime optimization for applying L update queries to an array of size N:

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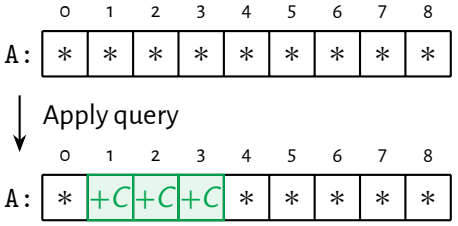
- Further optimizations

One-dimensional difference arrays



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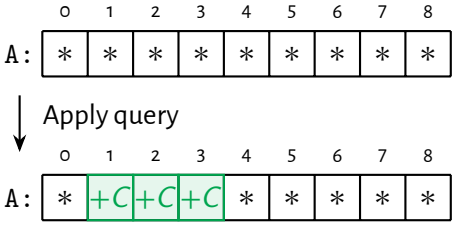

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"Mark" L queries in $\mathcal{O}(L)$ and perform the updates later in one sweep in $\mathcal{O}(N)$.

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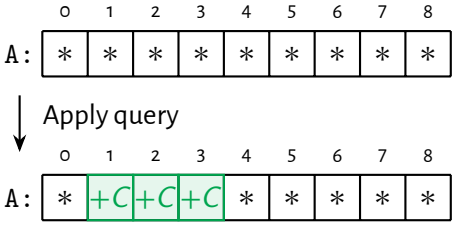

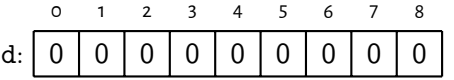

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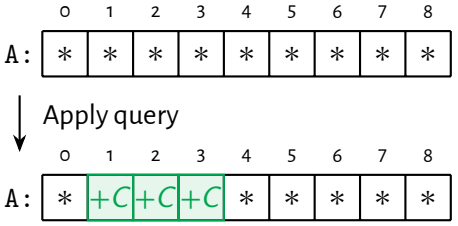

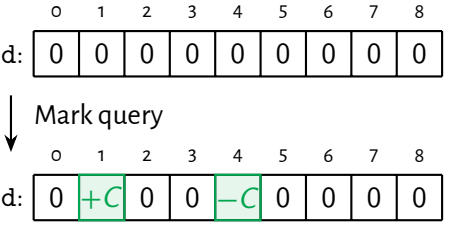

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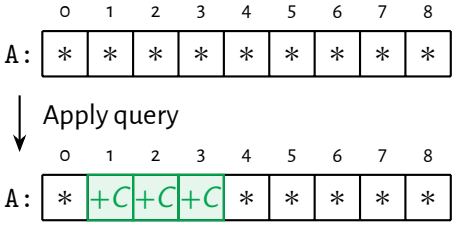

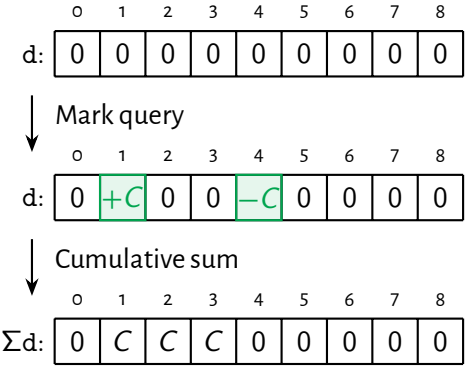

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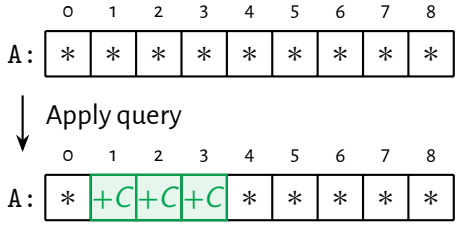

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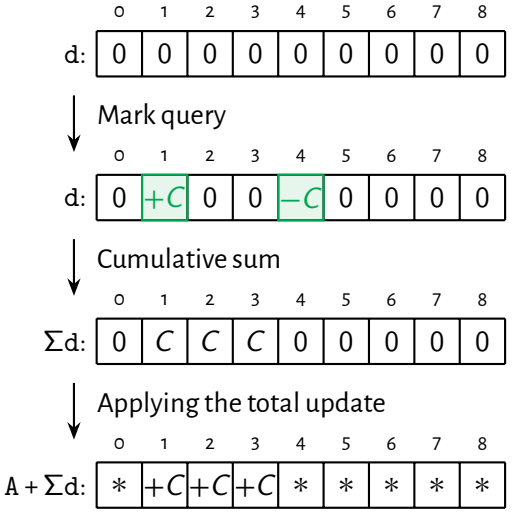

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For an array of size N and L queries of arbitrary size

Naive algorithm:

1. For every query add its update value over the whole update range

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1. Initialize array $\mathtt{d}$ of size N with zero entries
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Runtime: $\mathcal{O}(L + N)$

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- Imagine placing the markers as forming separate difference arrays $\mathtt{d}_1, \ldots, \mathtt{d}_L$, that is

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- The cumulative sum $\Sigma\mathtt{d}$ can be computed in $\mathcal{O}(N)$ by reusing previous results:

$$\Sigma\mathtt{d}[j] = \mathtt{d}[j] + \sum_{i=1}^{j-1} \mathtt{d}[i] = \mathtt{d}[j] + \Sigma\mathtt{d}[j-1].$$

Two-dimensional difference arrays

This concept can easily be generalized to 2D arrays.

*	*	*	*	*	*	*	*
*	$+C$	$+C$	$+C$	$+C$	$+C$	*	*
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	$-C$					$+C$	
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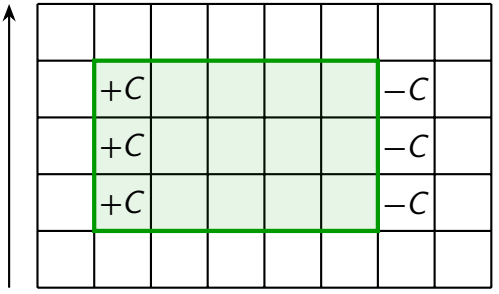
Placing *four* markers in the difference array.

Two-dimensional difference arrays

Taking cumulative sums in both directions yields the correct upates.



Cumulative sum along the y-axis

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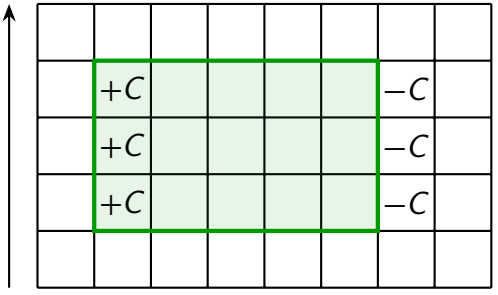

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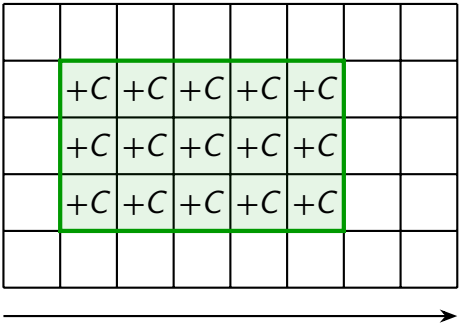

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- To leaverage these tools, we first need to formally describe our situation

The formal framework

We view arrays as maps over *finite grid domains*.

Definition (discrete maps, arrays):

- For $a \leq b \in \mathbb{Z}$ define the *discrete interval* $[\![a, b]\!] := \{m \in \mathbb{Z} \mid a \leq m \leq b\}$

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- A *discrete map* is a map $f : D \to \mathbb{R}$ over a finite grid domain $D \subseteq \mathbb{Z}^d$
- A *d-dimensional array* is a discrete map $f : \prod_{i=1}^{d} [\![0, m_i]\!] \to \mathbb{R}$

Definition (constant update queries):

- A discrete map $q : D \to \mathbb{R}$ which is constant on the *update range* $D_q = \prod_{i=1}^{d} [\![a_i, b_i]\!] \subseteq D$ is called *constant update query*

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Let $f : D \to \mathbb{R}$ with $D = \prod_{i=1}^{d} [\![n_i, m_i]\!]$ be a discrete function and $q_1, \dots, q_L$ a series of constant update queries.

Applying the series of queries to f means forming the updated discrete function $F : D \to \mathbb{R}$ as

$$F := f + \sum_{i=1}^{L} q_i.$$

A naive algorithm

Input: A discrete map $f : D \to \mathbb{R}$ with $D = \prod_{i=1}^{d} [\![n_i, m_i]\!]$ and a series of constant update queries $q_1, \dots, q_L$ on f over the update ranges $D_j = \prod_{i=1}^{d} \left[\!\left[a_i^{(j)}, b_i^{(j)}\right]\!\right]$ with update values C_j for $1 \leq j \leq L$

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1: $F \leftarrow$ The map $f : D \to \mathbb{R}$
2: **for** $j \in [\![1, L]\!]$ **do**
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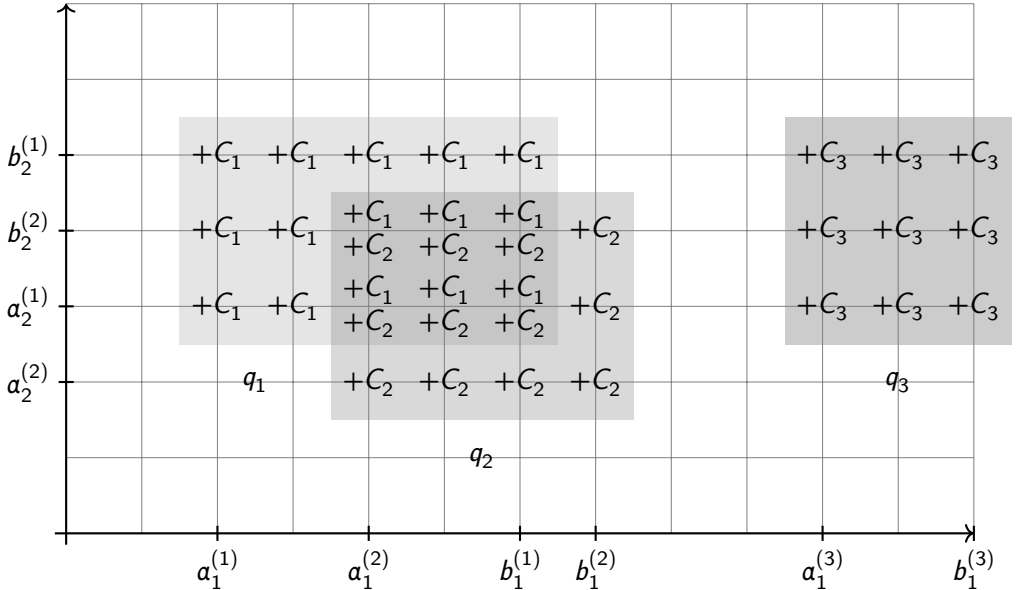
Runtime: $\mathcal{O}(L \cdot N)$ with $N := \mathrm{card}(D)$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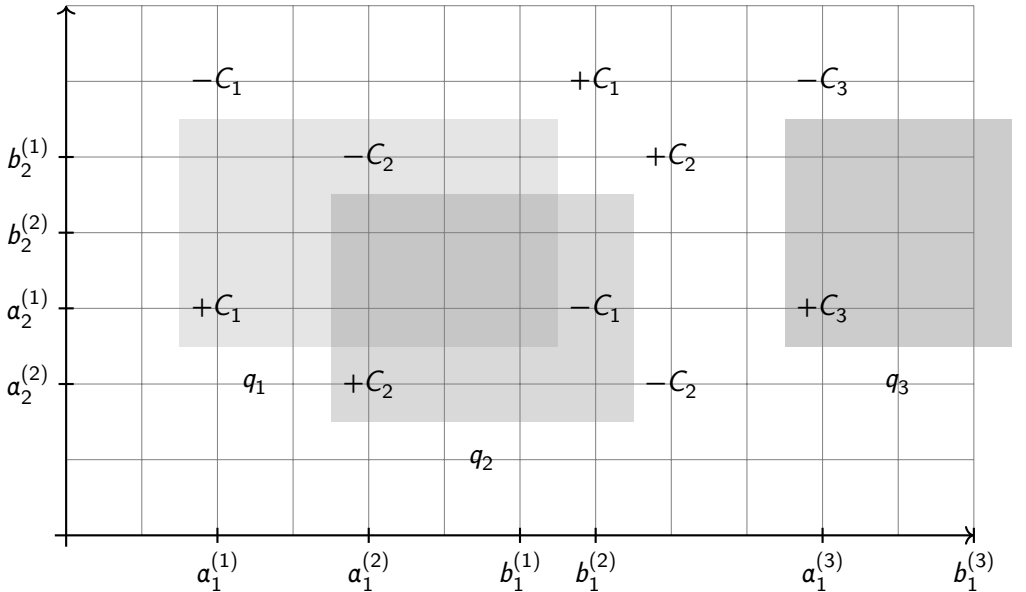
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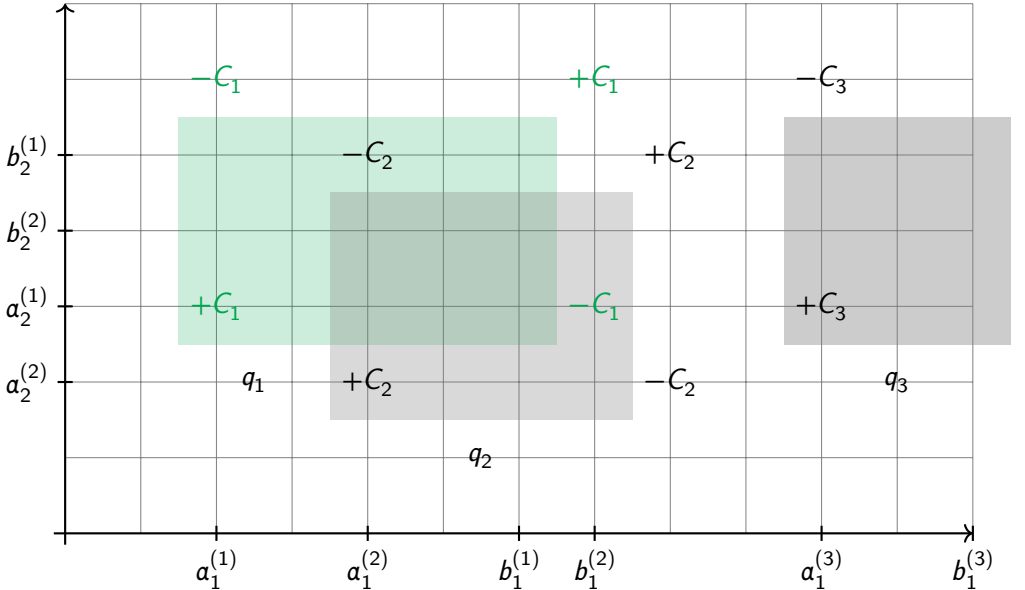
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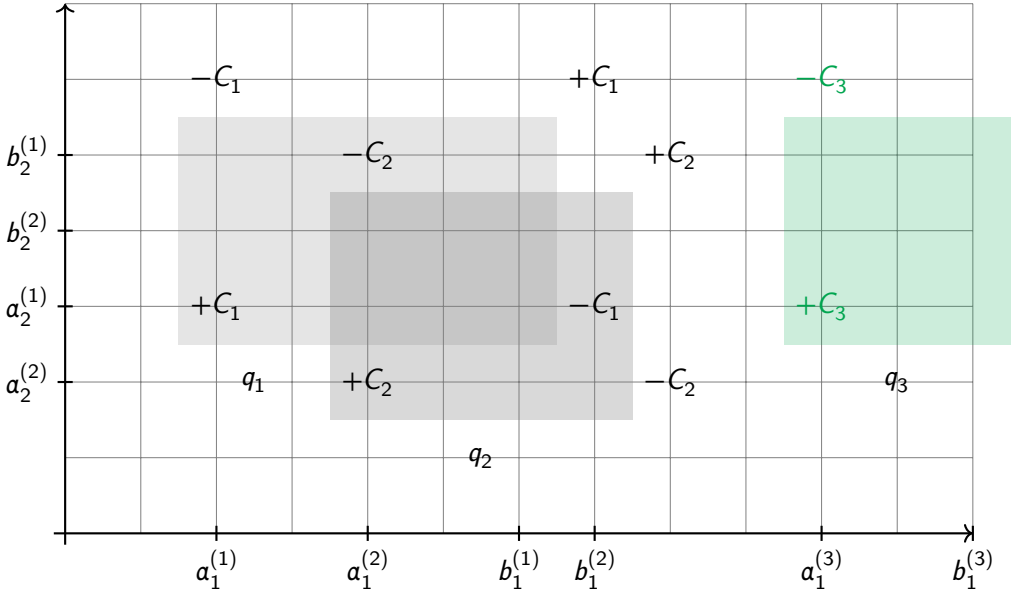
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The notation δ_p means the Kronecker-delta.

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- $q_1, \dots, q_L$ be a series of constant update queries on D over the update ranges $D_j = \prod_{i=1}^{d} [\![a_i^{(j)}, b_i^{(j)}]\!]$ with update values C_j for $j \in \{1, \dots, L\}$,

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The *difference array* $\Delta[q_1, \dots, q_L] : D \to \mathbb{R}$ is defined as

$$\Delta[q_1, \dots, q_L] = \sum_{j=1}^{L} \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p,$$

where $N_j(p) = \mathrm{card}\{p_i \mid p_i = b_i^{(j)} + 1\}$ for $j \in \{1, \dots, L\}$.

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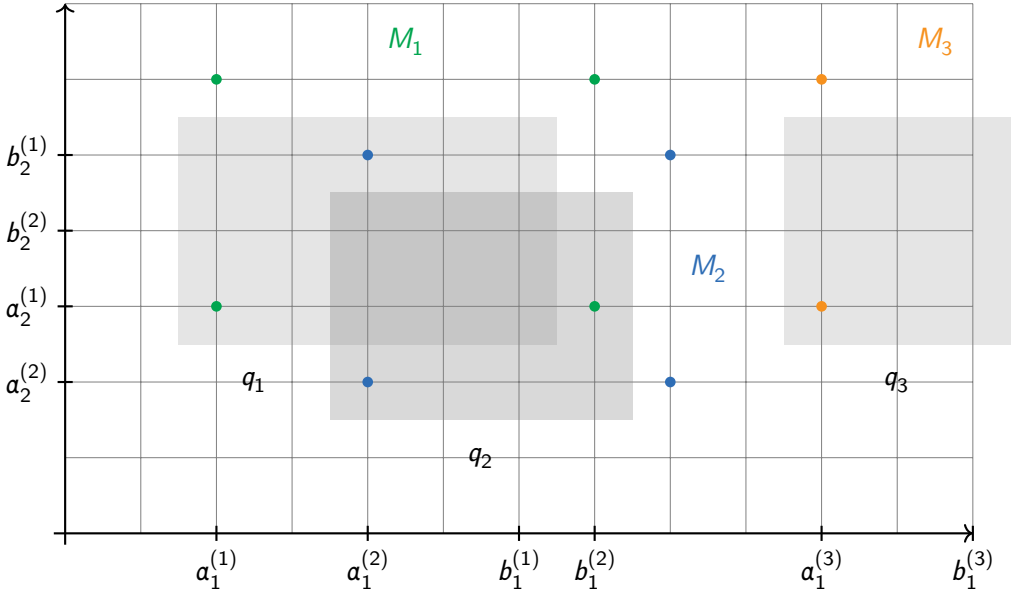
M_1
M_2
M_3
$b_2^{(1)}$
$b_2^{(2)}$
$a_2^{(1)}$
$a_2^{(2)}$
q_1
q_2
q_3
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Procedure: DIFFARR$(q_1, \dots, q_L)$

Input: A finite grid domain $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ and a series of constant update queries $q_1, \dots, q_L$ on D over the update ranges $D_j = \prod_{i=1}^{d} [\![a_i^{(j)}, b_i^{(j)}]\!]$ with update values C_j for $1 \leq j \leq L$

Output: The difference array $\Delta[q_1, \dots, q_L] = \sum_{j=1}^{L} \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p$

1: $d \leftarrow$ Zero map $d : D \to \mathbb{R}$
2: **for** $j \in \{1, \dots, L\}$ **do**
3: $\quad M_j \leftarrow \prod_{i=1}^{d} \left(\left\{ a_i^{(j)} \right\} \cup \left\{ b_i^{(j)} + 1 \mid b_i^{(j)} \neq m_i \right\} \right)$
4: $\quad$ **for** $p \in M_j$ **do**
5: $\quad\quad N(p) \leftarrow \mathrm{card}\{p_i \mid p_i = b_i^{(j)} + 1\}$
6: $\quad\quad d(p) \leftarrow d(p) + (-1)^{N(p)} \cdot C_j$
7: $\quad$ **end for**
8: **end for**
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7: $\quad$ **end for**
8: **end for**
9: **return** d

Runtime: The runtime of the body of the outer for-loop is *constant*!
$\Longrightarrow$ Total runtime: $\mathcal{O}(N + L)$ with $N := \mathrm{card}(D)$

Towards Discrete Calculus

How does this formal notion of a difference array help us in computing the updated map $F = f + \sum_{i=1}^{L} q_i$ more efficiently?

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From the 1D/2D heuristic: Cumulatively summing over $\Delta[q_1, \ldots, q_L]$ should yield the total update $\sum_{i=1}^{L} q_i$.

Roadmap: How does Discrete Calculus come into play?

- We will see that a difference array $\Delta[q_1, \dots, q_L]$ is the sum of discrete derivatives of the queries $q_1, \dots, q_L$, that is

$$\Delta[q_1, \dots, q_L] = \sum_{i=0}^{L} \Delta q_i.$$

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- Cumulatively summing over $\Delta[q_1, \dots, q_L]$ will be the discrete analogon of integration and thus will be the inverse operation to discrete differentiation, that is

$$\Sigma(\Delta[q_1, \dots, q_L]) = \sum_{i=0}^{L} \Sigma(\Delta q_i) = \sum_{i=0}^{L} q_i.$$

Foundations of Discrete Calculus

Definition: The *i-th backward difference* Δ_i is the linear operator assigning to every discrete map $f : D \subseteq \mathbb{Z}^d \to \mathbb{R}$ with $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ the map $\Delta_i f : D \to \mathbb{R}$ given by

$$\begin{aligned}\Delta_i f(k_1, \dots, k_d) := f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d) \\ - f(k_1, \dots, k_{i-1}, k_i - 1, k_{i+1}, \dots, k_d)\end{aligned}$$

for all $(k_1, \dots, k_d) \in D$ with $k_i > n_i$ and

$$\Delta_i f(k_1, \dots, k_d) := f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d)$$

for all $(k_1, \dots, k_d) \in D$ with $k_i = n_i$.

Definition (continued): For shorthand notation, we use

$$\Delta := \Delta_1 \circ \cdots \circ \Delta_d$$

to denote the *mixed backward difference* Δ.

Definition: The *i-th prefix sum* Σ_i is the linear operator assigning to every discrete function $f : D \subseteq \mathbb{Z}^d \to \mathbb{R}$ with $D = \prod_{j=1}^{d} [\![n_i, m_i]\!]$ the function $\Sigma_i f : D \to \mathbb{R}$ given by

$$\Sigma_i f(k_1, \dots, k_d) := \sum_{t=n_i}^{k_i} f(k_1, \dots, k_{i-1}, t, k_{i+1}, \dots, k_d)$$

for all $(k_1, \dots, k_d) \in D$. Again for shorthand notation, we use

$$\Sigma := \Sigma_1 \circ \cdots \circ \Sigma_d$$

to denote the *mixed prefix sum* Σ.

Properties of backward differences and prefix sums

- It is not hard to show that backward differences and prefix sums commute, that is for every i and j we have $\Sigma_i \circ \Delta_j = \Delta_j \circ \Sigma_i$ and thus

$$\Sigma \circ \Delta = (\Sigma_1 \circ \Delta_1) \circ \cdots \circ (\Sigma_d \circ \Delta_d) = \Delta \circ \Sigma.$$

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- The i-th backward difference Δ_i and the i-th prefix sum Σ_i are inverse to each other for all i, that is for every discrete map f we have $\Delta_i \circ \Sigma_i(f) = f$ and thus
$$\Delta \circ \Sigma(f) = f = \Sigma \circ \Delta(f).$$

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- Prefix sums can be computed efficiently:

$$\begin{aligned}
&\Sigma_i(f)(k_1, \dots, k_d) \\
&\quad = f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d) + \sum_{t=n_i}^{k_i-1} f(k_1, \dots, k_{i-1}, t, k_{i+1}, \dots, k_d) \\
&\quad = f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d) + \Sigma_i(f)(k_1, \dots, k_{i-1}, k_i - 1, k_{i+1}, \dots, k_d).
\end{aligned}$$

Procedure: PrefixSum(f, j)

Input: A discrete map $f : D \to \mathbb{R}$ with $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ and an index $j \in \{1, \dots, d\}$.

Output: The j-th prefix sum $\Sigma_j(f) : D \to \mathbb{R}$

1: $S \leftarrow$ The discrete map $f : D \to \mathbb{R}$
2: **for** $(k_1, \dots, k_{j-1}, k_{j+1}, \dots, k_d) \in D_{-j} := \prod_{\substack{i=1 \\ i \neq j}}^{d} [\![n_i, m_i]\!]$ **do**
3: **for** $t \in [\![n_j + 1, m_j]\!]$ **do**
4: $S(k_1, \dots, k_{j-1}, t, k_{j+1}, \dots, k_d) \leftarrow S(k_1, \dots, k_{j-1}, t-1, k_{j+1}, \dots, k_d) + S(k_1, \dots, k_{j-1}, t, k_{j+1}, \dots, k_d)$
5: **end for**
6: **end for**
7: **return** S

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5: **end for**
6: **end for**
7: **return** S

Runtime: The runtime of the outer for-loop lies in $\mathcal{O}\left(\text{card}(D_{-j}) \cdot \text{card}([\![n_j + 1, m_j]\!])\right) = \mathcal{O}(\text{card}(D))$
$\Longrightarrow$ Total runtime: $\mathcal{O}(N)$ with $N = \text{card}(D)$.

Procedure: MixedPrefixSum(f)
Input: A discrete map $f : D \to \mathbb{R}$ with $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$
Output: The mixed prefix sum $\Sigma(f) : D \to \mathbb{R}$

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2: **for** $j \in \{1, \dots, d\}$ **do**
3: $\quad S \leftarrow$ PrefixSum(S, j)
4: **end for**
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2: **for** $j \in \{1, \dots, d\}$ **do**
3: $\quad S \leftarrow$ PrefixSum(S, j)
4: **end for**
5: **return** S

Runtime: $\mathcal{O}(N)$ with $N = \mathrm{card}(D)$

Indicator functions and Heaviside step functions

Definition: Let $p = (p_1, \dots, p_d) \in \mathbb{Z}^d$. The *Heaviside step function* $H_p : \mathbb{Z}^d \to \mathbb{R}$ is the indicator function given by

$$H_p(k_1, \dots, k_d) := \begin{cases} 1, & \text{if } \forall 1 \leq i \leq d : k_i \geq p_i, \\ 0, & \text{else} \end{cases}$$

for $(k_1, \dots, k_d) \in \mathbb{Z}^d$.

Heaviside step functions are specific indicator functions with very simple mixed backward differences.

Proposition: Let $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ be a finite grid domain and $p = (p_1, \dots, p_d) \in D$. The mixed backward difference ΔH_p of the Heaviside step function $H_p : D \to \mathbb{R}$ is given by

$$\Delta H_p(k_1, \dots, k_d) = \delta_p := \begin{cases} 1, & \text{if } \forall 1 \leq i \leq d : k_i = p_i, \\ 0, & \text{else.} \end{cases}$$

Proposition: Let $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ be a finite grid domain and $p = (p_1, \dots, p_d) \in D$. The mixed backward difference ΔH_p of the Heaviside step function $H_p : D \to \mathbb{R}$ is given by

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Proof.

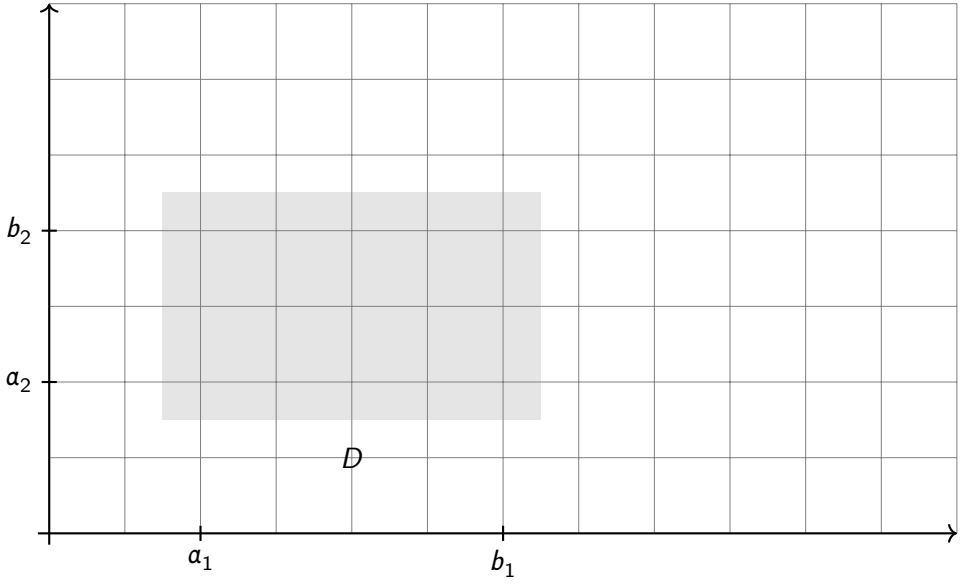
- It is straight forward to see that $H_p = \prod_{i=1}^{d} H_{p_i}$
- One can show that for such products we have $\Delta H_p = \prod_{i=1}^{d} \Delta H_{p_i}$
- Show $\Delta H_{p_i} = \delta_{p_i}$ by a simple case distinction
- Finally, $\Delta H_p = \prod_{i=1}^{d} \delta_{p_i} = \delta_{(p_1,\dots,p_d)}$

□

Proposition: Let $D = \prod_{i=1}^{d} [\![a_i, b_i]\!] \subseteq \mathbb{Z}^d$ be a finite grid domain. Then, the indicator function $\mathbb{1}_D : \mathbb{Z}^d \to \mathbb{R}$ can be written as

$$\mathbb{1}_D = \sum_{p \in \prod_{i=1}^{d} \{a_i, b_i+1\}} (-1)^{N(p)} H_p,$$

where $N(p) = \text{card}\{p_i \mid p_i = b_i + 1\}$.


b_2
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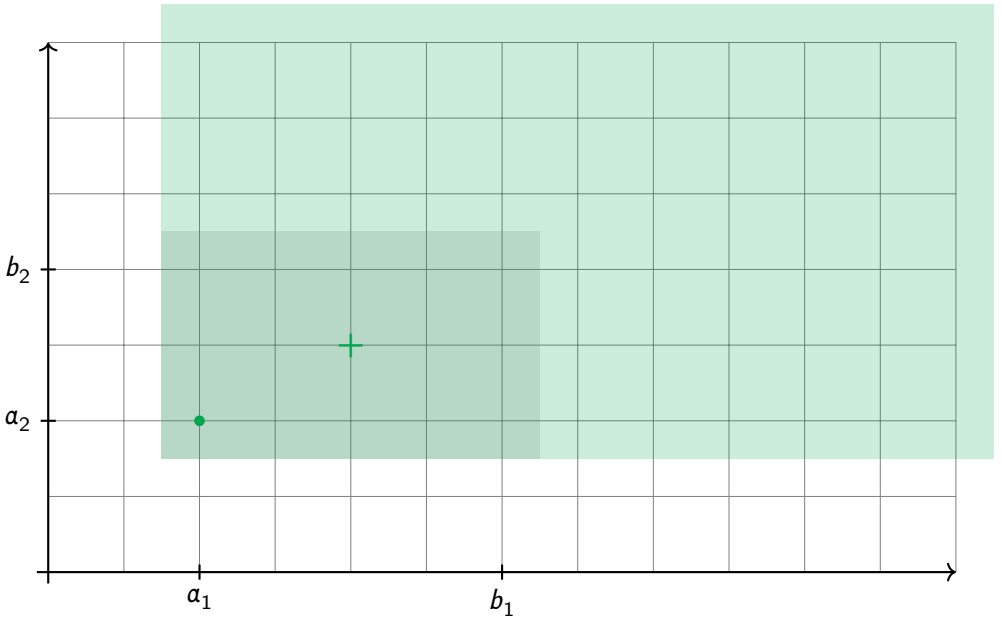
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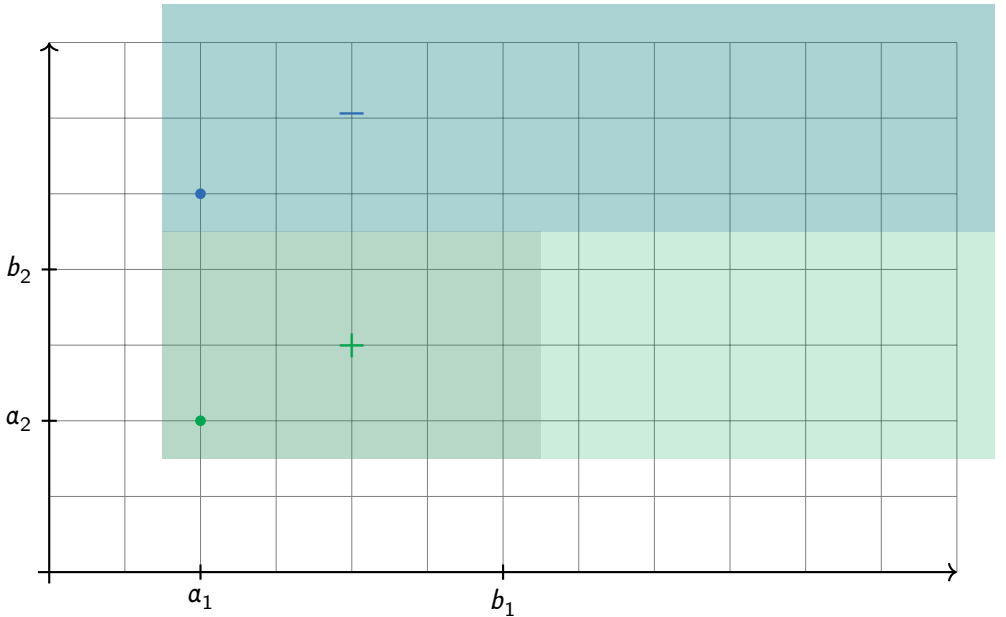
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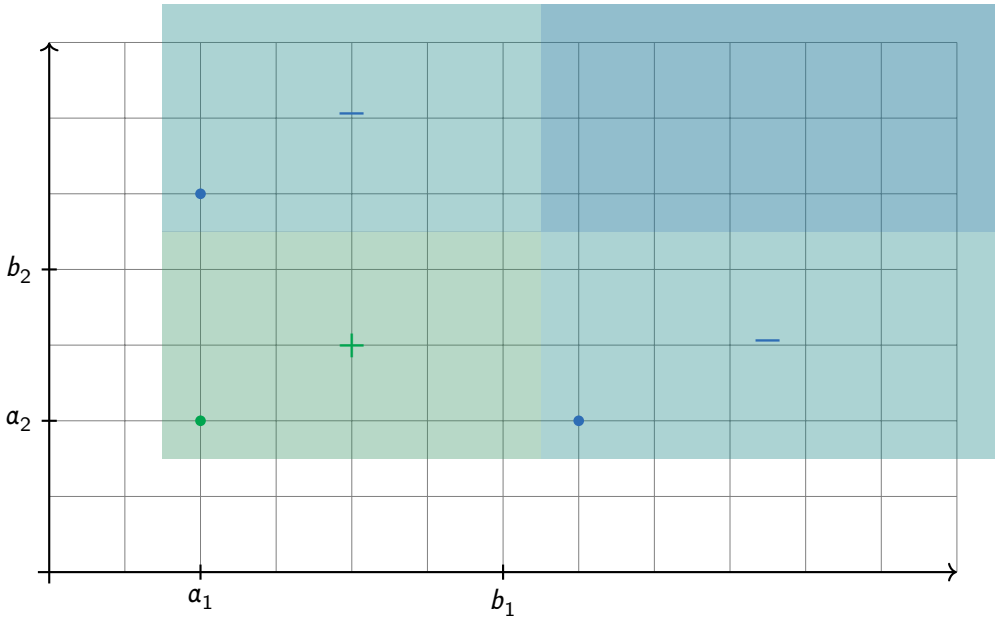
b_2
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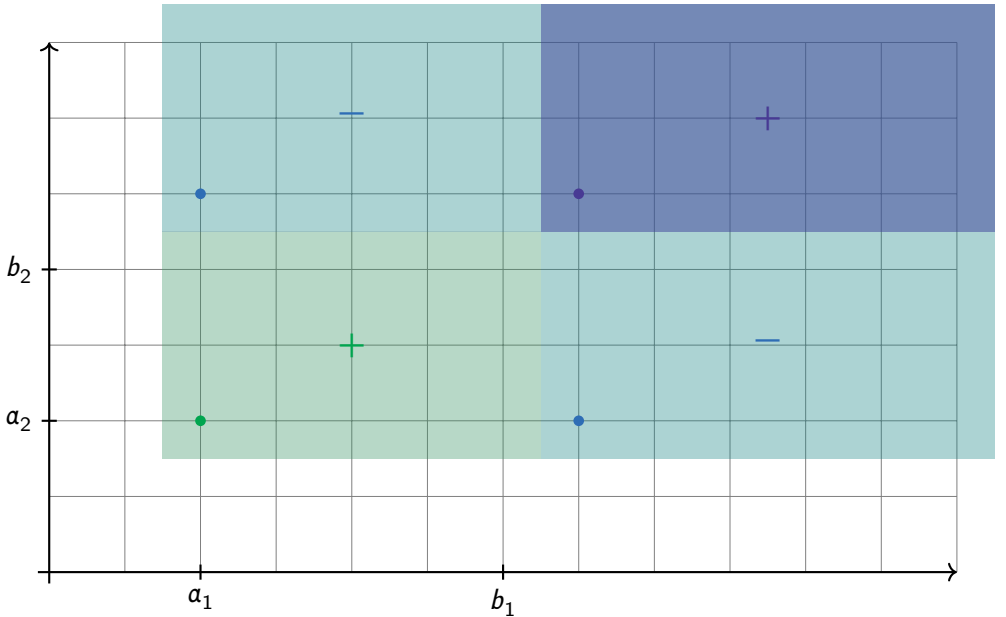
b_2
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$$\mathbb{1}_D = \sum_{p \in \prod_{i=1}^{d} \{a_i, b_i+1\}} (-1)^{N(p)} H_p,$$

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Proof.

- It is not hard to see that $\mathbb{1}_D(k_1, \dots, k_d) = \prod_{i=1}^{d} \mathbb{1}_{[\![a_i, b_i]\!]}(k_i)$
- By a simple case distiction we see that $\mathbb{1}_{[\![a_i, b_i]\!]}(k_i) = H_{a_i}(k_i) - H_{b_i+1}(k_i)$
- Hence, $\mathbb{1}_D(k_1, \dots, k_d) = \prod_{i=1}^{d} \left(H_{a_i}(k_i) - H_{b_i+1}(k_i) \right)$
- Using the general distribution formula for products yields

$$\prod_{i=1}^{d} \left(H_{a_i}(k_i) - H_{b_i+1}(k_i) \right) = \sum_{J \subseteq I} (-1)^{\text{card}(J)} \left(\prod_{i \in I \setminus J} H_{a_i}(k_i) \right) \left(\prod_{j \in J} H_{b_j+1}(k_j) \right),$$

where $I = \{1, \dots, d\}$.

Proof (continued). ▪ Then for every $J \subseteq I$ we have

$$\left(\prod_{i \in I \setminus J} H_{a_i}(k_i) \right) \left(\prod_{j \in J} H_{b_j+1}(k_j) \right) = H_p(k_1, \dots, k_d)$$

with $p = (p_1, \dots, p_d)$ given by

$$p_i = \begin{cases} a_i, & \text{if } i \in I \setminus J, \\ b_i + 1, & \text{if } i \in J. \end{cases}$$

▪ These are exactly the points in $\prod_{i=1}^{d} \{a_i, b_i + 1\}$, hence it is not hard to verify that indeed

$$\sum_{J \subseteq I} (-1)^{\operatorname{card}(J)} \left(\prod_{i \in I \setminus J} H_{a_i}(k_i) \right) \left(\prod_{j \in J} H_{b_j+1}(k_j) \right) = \sum_{p \in \prod_{i=1}^{d} \{a_i, b_i+1\}} (-1)^{N(p)} H_p(k_1, \dots, k_d)$$

with $N(p) = \operatorname{card}\{p_i \mid p_i = b_i + 1\}$. □

Corollary: Let $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ be a finite grid domain and let $q : D \to \mathbb{R}$ be a constant update query on D with update range $D_q = \prod_{i=1}^{d} [\![a_i, b_i]\!]$ and update value $C \in \mathbb{R}$. Then,

$$q = \sum_{p \in M} (-1)^{N(p)} \cdot C \cdot H_p,$$

where $M = \prod_{i=1}^{d} (\{a_i\} \cup \{b_i + 1 \mid b_i \neq m_i\})$ and $N(p) = \operatorname{card}\{p_i \mid p_i = b_i + 1\}$. In particular, the mixed backward difference Δq of q can be expressed as

$$\Delta q = \sum_{p \in M} (-1)^{N(p)} \cdot C \cdot \delta_p.$$

Proof. ▪ By definition $q = C \cdot \mathbb{1}_{D_q}$

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Proof.

- By definition $q = C \cdot \mathbb{1}_{D_q}$
- By the previous proposition:

$$q = \sum_{p \in \prod_{i=1}^{d} \{a_i, b_i + 1\}} (-1)^{N(p)} \cdot C \cdot H_p.$$

Proof (continued). ▪ We are not viewing q as a function over $\mathbb{Z}^d$ but over $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$

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- If $p \in \prod_{i=1}^{d} \{a_i, b_i + 1\}$ with $p_j = b_j + 1$ and $b_j = m_j$, then $p \notin D$

Proof (continued).

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Proof (continued).

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- Lastly, the form of Δq follows directly by the linearity of the mixed backward difference Δ and $\Delta H_p = \delta_p$. □

Recall the formal definition of difference arrays:

Definition: Let $D = \prod_{i=1}^{d} [\![n_i, m_i]\!] \subseteq \mathbb{Z}^d$ be a finite grid domain and $q_1, \dots, q_L$ a series of constant update queries on D. The *difference array* $\Delta[q_1, \dots, q_L] : D \to \mathbb{R}$ is defined as

$$\Delta[q_1, \dots, q_L] = \sum_{j=1}^{L} \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p,$$

where $M_j = \prod_{i=1}^{d} \left(\left\{ a_i^{(j)} \right\} \cup \left\{ b_i^{(j)} + 1 \mid b_i^{(j)} \neq m_i \right\} \right)$ and $N_j(p) = \text{card}\{p_i \mid p_i = b_i^{(j)} + 1\}$ for $1 \leq j \leq L$.

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- With the corollary on the last slide we see that

$$\Delta q_j = \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p$$

for all $j \in \{1, \dots, L\}$.

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$$\Delta[q_1, \dots, q_L] = \sum_{j=1}^{L} \Delta q_j.$$

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- We have seen that the prefix sum $\Sigma(\Delta[q_1, \dots, q_L])$ can be computed in $\mathcal{O}(N)$.
- The addition $f + \Sigma(\Delta[q_1, \dots, q_l])$ has time complexity $\mathcal{O}(N)$.
- In total: Computing $f + \Sigma(\Delta[q_1, \dots, q_l])$ can be done in $\mathcal{O}(L + N)$.

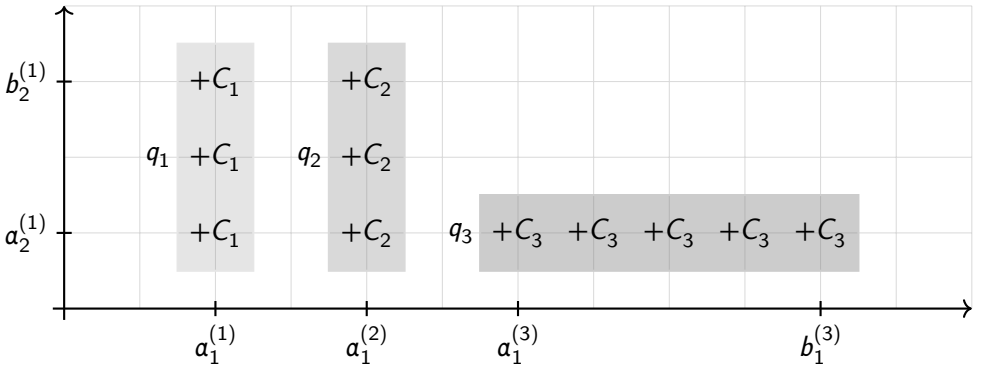
Input: A discrete map $f : D \to \mathbb{R}$ with $D = \prod_{i=1}^{d} [\![n_i, m_i]\!]$ and a series of constant update queries $q_1, \dots, q_L$ on D

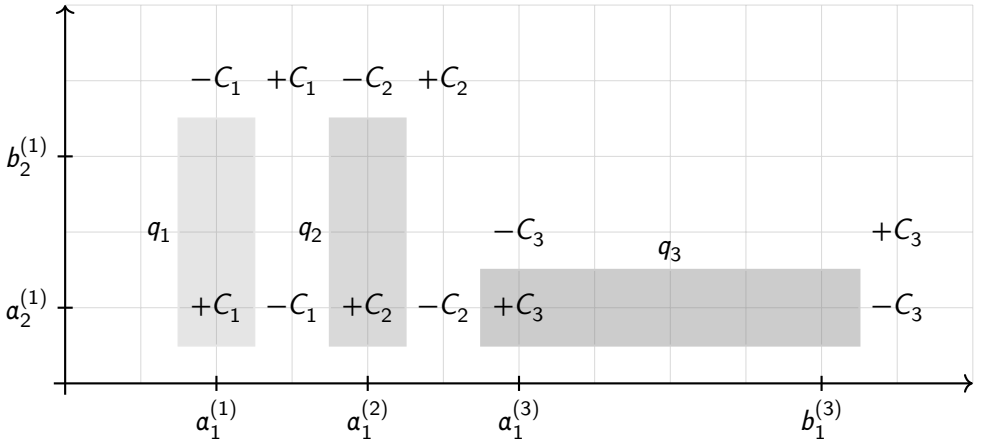
Output: The updated map $F = f + \sum_{i=1}^{L} q_i$

1: $d \leftarrow \textsc{DiffArr}(D, q_1, \dots, q_L)$
2: $\Sigma d \leftarrow \textsc{MixedPrefixSum}(d)$
3: $F \leftarrow f + \Sigma d$
4: **return** F

Runtime: $\mathcal{O}(L+N+N+N) = \mathcal{O}(L+N)$
with $N := \mathrm{card}(D)$

Flat update queries




$-C_1$
$+C_1$
$-C_2$
$+C_2$
$b_2^{(1)}$
q_1
q_2
$-C_3$
q_3
$+C_3$
$a_2^{(1)}$
$+C_1$
$-C_1$
$+C_2$
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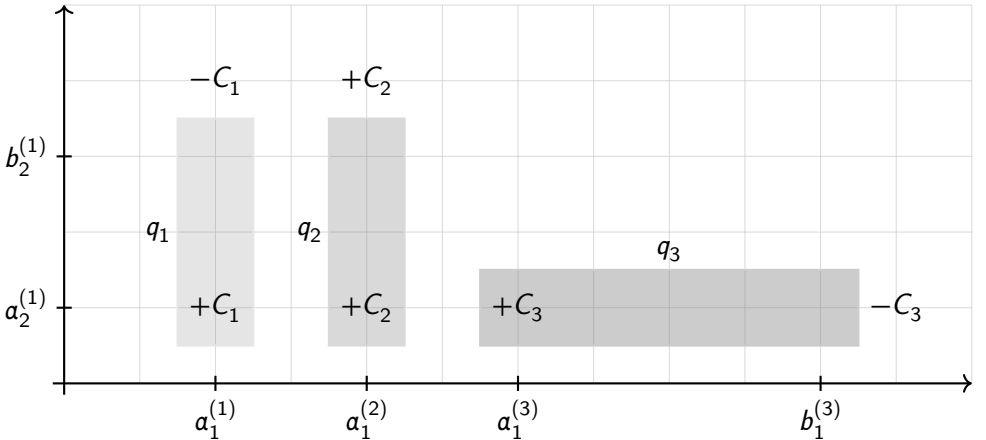
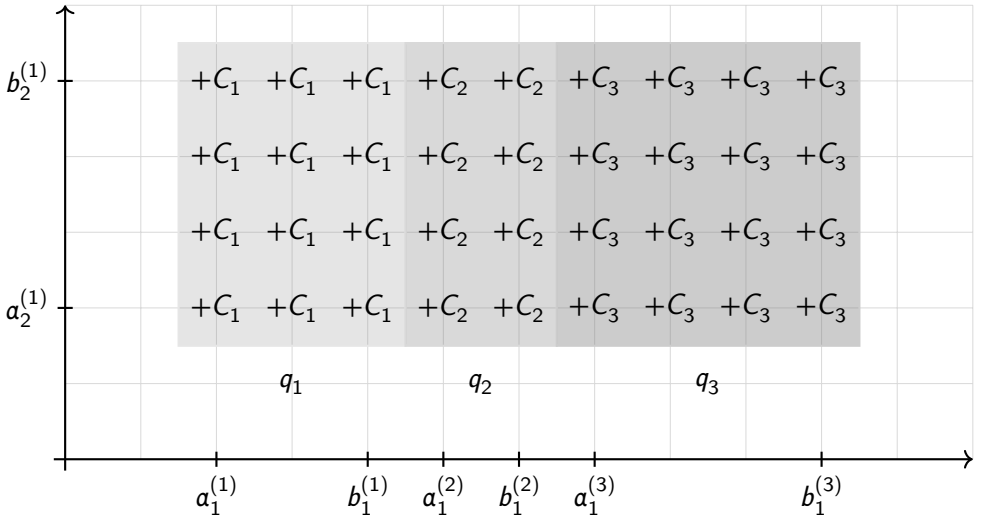
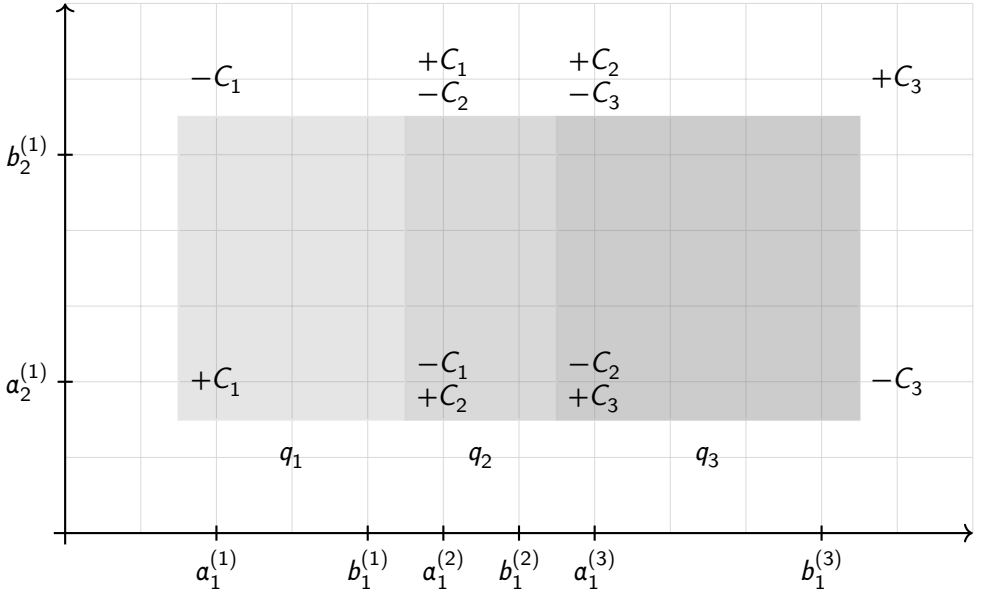
$+C_3$
$-C_3$
$a_1^{(1)}$
$a_1^{(2)}$
$a_1^{(3)}$
$b_1^{(3)}$


$-C_1$
$+C_2$
$b_2^{(1)}$
q_1
q_2
q_3
$a_2^{(1)}$
$+C_1$
$+C_2$
$+C_3$
$-C_3$
$a_1^{(1)}$
$a_1^{(2)}$
$a_1^{(3)}$
$b_1^{(3)}$

Consecutive update queries

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q_1
q_2
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Any questions?

(`https://github.com/mazmap/multidimensional-difference-arrays`)

↑ Slides ↑